EVAN BONERTZ

Lead Software Engineer / Senior GenAI Engineer

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SUMMARY

Highly versatile Lead Software Engineer and Full-stack Developer with over a decade of experience delivering enterprise-ready web, cloud, and AI/ML solutions. Proven track record of designing scalable cloud infrastructures and integrating production-grade GenAI pipelines to solve complex business problems in regulated spaces like healthcare, fintech, and insurance. Expert in TypeScript, Python, Node.js, and AWS/GCP, with a deep focus on LLM API integrations, prompt engineering, and automated workflows.

SKILLS & TECHNOLOGIES

- **AI / GenAI / ML:** LLM Integration & Evaluation, Prompt Engineering, RAG (Retrieval-Augmented Generation) Architecture, Vector Embeddings, Agentic Workflows & Tool Use (Function Calling), Python (Scikit-Learn, PyTorch, Hugging Face, OpenAI APIs)
- **Cloud & DevOps:** AWS (S3, Lambda, ECS, EC2, Bedrock, SNS/SQS, CloudWatch), GCP (GKE, App Engine, Cloud Functions, Pub/Sub, Cloud SQL), Docker, Kubernetes, CI/CD (GitHub Actions), Terraform
- **Web Development & Frameworks:** Node.js, TypeScript, Python, PHP, Next.js, Nest.js, React, Vue (Vue 3), Laravel, REST APIs, GraphQL
- **Databases & Search:** SQL (MySQL, PostgreSQL), NoSQL (DynamoDB, MongoDB, Firestore), ElasticSearch, OpenSearch

FEATURED AI PROJECTS

- **Automated Semantic Search & RAG Pipeline** | *Python, LangChain, Vector Database, OpenAI*
  - Designed and deployed a custom Retrieval-Augmented Generation (RAG) system to extract, chunk, and embed unstructured text documents into a vector database for semantic search.
  - Implemented intelligent prompt engineering and context-window optimization to reduce LLM hallucination rates while generating automated summaries of dense, multi-page data.
- **AI-Driven Automated Newsletter Agent** | *TypeScript, Node.js, AWS Lambda, LLM APIs*
  - Built a serverless, agentic automated workflow that aggregates regional web data, filters it using vector embeddings, and leverages an LLM to draft highly contextual, localized, and styled community newsletter content.

PROFESSIONAL EXPERIENCE

Full-stack Software Engineer II

November 2023 – Present

ISC | Raleigh, NC

- **AI Underwriting & Analysis:** Designed and built secure, robust REST APIs and databases supporting an enterprise AI underwriting platform (Lance Insights).
- **GenAI Quote Comparison:** Architected and deployed an AI Quote Comparison tool that processes complex, unstructured quote documents, leveraging LLMs to extract critical data points and generate actionable summaries for underwriters.
- **API Integrations:** Designed and implemented a custom API integration connecting an AI-based Excel rating engine to programmatically generate real-time insurance quotes.
- **Tech Stack & DevOps:** Engineered serverless background workers and backend integrations using TypeScript, Node.js, and AWS Lambda; deployed scalable microservices to AWS ECS/EC2 using GitHub Actions CI/CD pipelines.
- **Leadership & Innovation:** Evaluated, optimized, and presented corporate-wide GitHub Copilot best practices to the ISCX developer team. Mentored junior engineers and led agile sprint planning to accelerate team velocity.

Lead Software Engineer

September 2021 – September 2023

Boomy | Raleigh, NC

- **Engineering Leadership:** Directed an agile team of 4 software developers to build, scale, and maintain a highly viral generative AI Music platform, contributing to 100% growth in Monthly Recurring Revenue (MRR).
- **Cloud Infrastructure:** Architected high-performance custom API integrations, message workers, and serverless background tasks across GCP (App Engine, GKE, Cloud Functions) and AWS (Lambda).
- **Full-Stack & Payments:** Developed and maintained subscriber-facing React/Vue web applications, leveraging Stripe for subscription management and structured feature flags for safe rolling deployments.
- **Developer Experience:** Standardized API documentation via Swagger and Postman; containerized the complete microservices suite using Docker to streamline local development.

Professional Services Engineer

March 2020 – September 2021

Integrify | Raleigh, NC

- **Enterprise Integration:** Collaborated directly with Fortune 500 engineering leaders to architect, build, and deploy custom workflow automation solutions utilizing Node.js and AWS Lambda.
- **Infrastructure & Alerting:** Designed and implemented reliable, automated error-alerting and monitoring systems utilizing AWS CloudWatch.

Cloud Engineer

June 2016 – March 2020

commercetools | Durham, NC

- **Cloud Native Solutions:** Developed and deployed enterprise SaaS APIs using GCP, AWS, Docker, and Kubernetes for global clients including Hasbro, Express, and Lego.

EDUCATION

- **B.S. Economics, B.S. Management and Society** | University of North Carolina at Chapel Hill
- **Front End Engineering Certificate** | The Iron Yard Academy